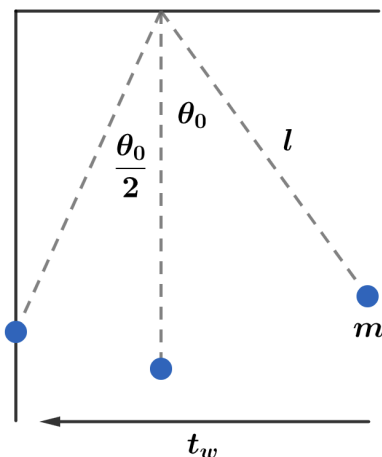


2023 F=ma Exam: Problem 16

Kevin S. Huang



The bead is in simple harmonic motion since it is a simple pendulum undergoing small oscillations. Thus, we have

$$\theta(t) = \theta_0 \cos(\omega t)$$

where

$$\omega = \sqrt{\frac{g}{l}}$$

is the angular frequency of a simple pendulum. Note $\theta(t = 0) = \theta_0$ as required. Then the time t_w it takes to reach the wall from the start is given by

$$\begin{aligned} -\frac{\theta_0}{2} &= \theta_0 \cos(\omega t_w) \\ \cos(\omega t_w) &= -\frac{1}{2} \\ \omega t_w &= \frac{2\pi}{3} \\ t_w &= \frac{2\pi}{3} \sqrt{\frac{l}{g}} \end{aligned}$$

This is half the time between wall collisions so

$$t_c = 2t_w = \frac{4\pi}{3} \sqrt{\frac{l}{g}} = \frac{4\pi}{3} \sqrt{\frac{10 \text{ m}}{10 \text{ m/s}^2}} = \frac{4\pi}{3} \text{ s}$$

so the answer is C.